

# ANUBHAV SHRIJAN MISHRA

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Kattankulathur, Chennai, Tamil Nadu 603203

## OBJECTIVE

Electronics and Communication Engineering undergraduate focused on VLSI, Embedded Systems and Semiconductor Technologies with hands-on experience in PCB Design, MEMS Research and Hardware Development.

## EDUCATION

**SRM Institute of Science and Technology (SRMIST)** **CGPA: 9.42 / 10**
B.Tech – Electronics and Communication Engineering (Data Science)

**Chinmaya Vidyalaya, Sector-V B.S. City, Bokaro, Jharkhand** **69.40%**
Class XII – CBSE, 2022

**Adiwasi High School, Mangrutikar** **88.80%**
Class X – JAC, 2020

## PUBLICATION

**Wearable Smart Face Mask Based on Functionalized Graphene for Real-Time Monitoring of Human Respiration**
IEEE Sensor Journal · Co-author with Dr. Avik Sett · Nanotechnology Domain · View Paper

- Fabricated smart face masks using curcumin and melamine-functionalized rGO coatings for real-time breath monitoring, achieving response times of 2s (cur-rGO) and 3.2s (m-rGO).

## TECHNICAL SKILLS

- **EDA & Simulation Tools:**Verilog, ModelSim, Proteus, EasyEDA, SEMulator3D, Ansys Icepack, TinkerCad
- **Embedded Systems:** Embedded C, ATmega328, 8051, UART, SPI, I2C, Arduino
- **Hardware & PCB Design:** PCB Design, Analog Circuits, Electronic Cooling, MEMS Fabrication
- **Programming Languages:** C, Python, SQL
- **Core Domains:** VLSI, Semiconductor Devices, Embedded Systems
- **Networking:** Junos OS, Routing, Switching, Cloud Networking

## EXPERIENCE

- **Intern – Drive Next Automotive Lab** — SRM Technologies Pvt Ltd 17 Sep 2026 – Present
  - Design a miniaturized ADAS power supply delivering up to 50 W from a 24 V truck battery with 97% efficiency, stable operation, minimal components, compact PCB footprint and compliance with automotive standards.
- **Advanced Embedded System Developer Virtual Internship** — Microchip Technology Inc. Jan – Mar 2026
  - Completed a 10-week AICTE EduSkills virtual internship on advanced PIC/AVR peripherals; CAN networks and rapid-prototyping applications using Curiosity Nano and MPLAB X IDE.
- **Structural Analysis Virtual Internship** — Ansys Oct – Dec 2025
  - Completed a 10-week AICTE–EduSkills virtual internship on structural simulation and analysis fundamentals using Ansys tools

## PROJECTS

**Multiple Access Scheme for Wired Network — SRMIST** 2026
ESP32 · W5500 Ethernet · CSMA/CD · UDP · Arduino

- Built a hardware CSMA/CD demonstrator using ESP32 nodes over W5500 Ethernet, implementing carrier sensing, collision detection, jam signaling, backoff and retransmission with OLED status display also documented protocol behavior at the Data Link Layer (IEEE 802.3).

**LM386-Based Audio Amplifier Circuit using Solar Panel** Mar 2026 – May 2026
SRMIST · Analog Circuit Design · EasyEDA

- Designed and tested a wireless audio transmission system using light as the carrier medium, integrating a solar panel as the power source and verifying signal integrity across the analog chain.

**Micro Thermopile for Infrared Sensing — IISc Bangalore** Feb 2026 – April 2026
MEMS Fabrication · SEMulator3D · LaTeX · Semiconductor Process Design

- Established a complete MEMS-based infrared thermopile sensor fabrication flow in collaboration with IISc Bangalore, designing and simulating the device structure using SEMulator3D and documenting deposition, lithography and etching process steps.

## CERTIFICATIONS

- **Computer Communication and Networks** Aug 2026
- **Design Verification with SystemVerilog/UVM** Oct 2025
- **8051 Microprocessor: Foundation of Microprocessors** — Intel Oct 2025
- **Verilog HDL Fundamentals** Sep 2024
- **PCB Design and Fabrication** Mar 2024

## MEMBERSHIP

**IEEE Electron Device Society (EDS) — SRMIST** May 2025 – Present
Senior Student Chair · Technical Member · IEEE Membership

- Led technical initiatives and coordinated 60+ member events, mentoring junior IEEE members and driving participation across the SRMIST EDS chapter.

## VOLUNTEERING

**Community Teaching Program — Saraswati Shishu Vidya Mandir, Jharkhand** Jun – Jul 2025
Mentored primary school students in academics as part of a community development initiative.

## ACHIEVEMENTS

- 3rd Position — Best Project Exhibition, SRMIST Jul 2024
- Best Anchor Award — Saraswati Shishu Vidya Mandir May 2021

## LANGUAGES

English (Professional) Hindi (Native) German (Basic)